Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII Sem End Semester Examination Nov/Dec 2024

Tissue Engineering [BME 4071]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A)	Distinguish and analyse the ligand driven and receptor driven processes involved in cellular signalling in skin. Must mention the cell signalling pathways associated with.	(5)
1B)	Compare different stem cell niche.	(2)
1C)	Illustrate briefly, the role of various transcription factors pertaining to the enriched NANOG expression of mouse embryonic stem cells.	(3)
2A)	Analyze with schematics the decellularization and re-cellularization process pertaining to liver.	(2)
2B)	Analyse the stages of isolation of mouse embryonic stem cells and justify the use of feeder cell layers during isolation process. How do you assess the potential of these isolated stem cells in vivo?	(4)
2C)	Analyse how shear stress and VEGF regulate blood vessel formation. Discuss Autologous chondrocyte Transplantation (ACT) process.	(4)
3A)	Chitosan is a biopolymer extracted from waste materials (exoskeleton of prawn). How would you use chitosan to fabricate interconnected homogeneous porous scaffolds? Write down your strategies in a brief and logical manner.	(5)
3B)	Susil intends to extract collagen for the fabrication of a composite matrix. What type of isolation technique would be appropriate? Describe the steps in brief and justify the choice. Will the orientation of the collagen fibres make any impact in the design of the composite matrix?	(3)
3C)	How do you avoid subjectivity while assessing porosity based on SEM micrographs of polymeric scaffolds. Suggest a process with proper logic.	(2)

4A)	Differentiate the process involved in rate zonal and isopycnic gradient centrifugation process. Why is FACs not considered an ideal process of selection when isolated cells are meant for clinical application? Justify your views logically.	(4)
4B)	Explain with proper reason: (a)Trypsin cannot be used to dissociate human embryonic stem cell colonies in vitro. (b)Culture condition (2D/3D) influences cellular behaviour.	(3)
4C)	Mention the major components of tissue culture medium with purpose.	(3)
5A)	Why does somatic and adult stem cells fail to counter the limitation of “Hay Flick number”?	(3)
5B)	What would be your strategy to select specifically chondrocyte cells from a mixed population of cells containing hepatocytes along with neural cells?	(3)
5C)	Compare selective adhesion and pre-plating methods of cell selection. Analyse the pros and cons of these methods.	(4)

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